

9- Von Neumann & Harvard Computers

There are two computer architectures, which are *different in the way of accessing memories*: von Neumann Architecture (also names “ Princeton Architecture”) and Harvard Architecture.

In a **Von Neumann architecture** , programs and data are stored in the same memory and managed by the same information-handling subsystem.

In the **Harvard architecture** , programs and data are stored and handled by different subsystems. This is the essential difference between these two architectures.

Harvard architecture uses physically separate memories for their instructions and data, requiring dedicated buses for each of them. Instructions and operands can therefore be fetched simultaneously.

In the original "Harvard computer", built in 1944 and after which the architecture is named, the program-handling task and the data-handling task were sufficiently different to result in two different storage technologies.

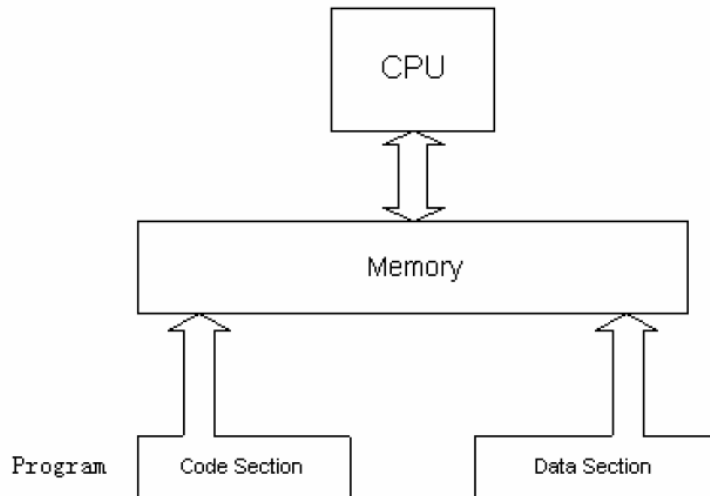
Today, the vast majority of computers are von Neumann architecture because of the efficiency of design, implementation, and operation of one memory system instead of two.

Under certain conditions, a Harvard computer can be much faster than a Von Neumann computer because data and program do not contend for the same information pathway, and storing the program in an immutable read-only memory can result in vast reliability improvements.

9-1 The von Neumann Architecture has following specialties :

1. Instructions and Data are stored in the same memory.
2. Instructions and Data share one memory system.

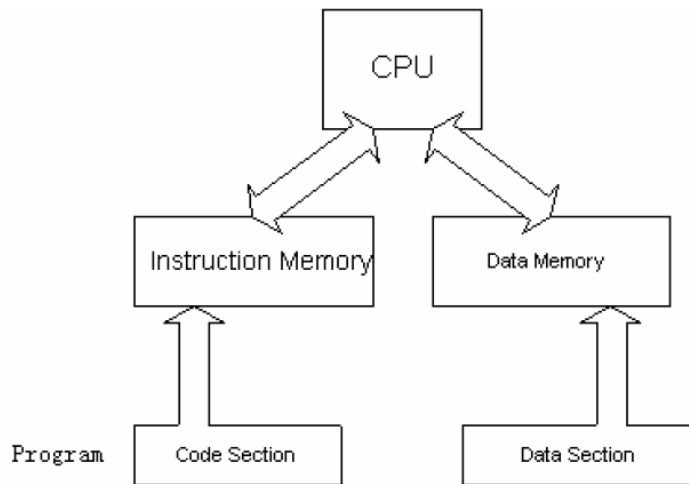
3. It has only one bus which is used for both data transfers and instruction fetches, and therefore data transfers and instruction fetches must be scheduled – they cannot be performed at the same time.



Fig(von Neumann architecture)

9-2 The Harvard Architecture has following specialties:

1. Physically separates storage and signal pathway for instructions and data.
2. It has separate data and instruction busses, allowing transfers simultaneously on both busses.
3. Generally, the bit of Instructions is wider than Data.
4. For some computers, the Instruction memory is read-only.
5. In cases without caches, the Harvard Architecture is more efficient than Von Neumann.



Fig(Harvard architecture)